Bellabeat Smart Wellness Project

A Data-Driven Analysis of Smart Device Usage to Identify Wellness Trends

Google Data Analytics Certificate Case Study

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Ask – Business Understanding

Business Context

Bellabeat is a women's health technology company founded in 2013 by Urška Sršen and Sandro Mur. The company designs smart wellness products that track activity, sleep, stress and reproductive health, with a focus on solutions built specifically for women.

Bellabeat has grown into a recognised wellness brand and has been featured in major publications such as Women's Health, Shape, Vogue, and Poosh. The company differentiates itself through jewellery-inspired smart devices, a women-first design philosophy and relatively affordable pricing compared with larger technology companies.

Business Task

Bellabeat’s marketing team wants to understand how consumers are using their smart devices. By analysing smart device usage data, the objective is to identify trends and patterns in user behaviour. These insights will help uncover potential growth opportunities and provide high-level recommendations to guide the marketing strategy of Bellabeat.

Key Business Questions

1. What are the trends in smart device usage among fitness tracker users?
2. How can these trends be applied to Bellabeat customers?
3. How can these insights influence Bellabeat's marketing strategy?

Stakeholders

The key stakeholders involved in this analysis include:

- Urška Sršen – Co-founder and Chief Creative Officer
- Sandro Mur – Co-founder and key executive team member
- Bellabeat Marketing Analytics Team – Responsible for analyzing user data and developing marketing strategies
- Bellabeat Executive Team – Decision-makers responsible for business growth and strategic direction

Prepare – Data Source and Description

Data Source

Dataset: FitBit Fitness Tracker Data (Kaggle, CC0 Public Domain)

The dataset contains personal fitness tracker data from 35 consenting users, covering daily activity, steps, calories, sleep and heart rate metrics. Since Fitbit offers functionality similar to Bellabeat products, this dataset is suitable for analyzing smart device usage trends.

Dataset Overview

Key Dataset Details

- Source: Kaggle – FitBit Fitness Tracker Data
- Licence: CC0 Public Domain (open for use)
- Time Period: March 2016 to May 2016
- Users: 35 consenting Fitbit users

Files used: Daily activity, sleep and heart rate data

Data Credibility – ROCCC Assessment

- **Reliable:** Moderate – small sample size; may not represent all Bellabeat users
- **Original:** Yes – data collected directly from Fitbit devices
- **Comprehensive:** Moderate – includes activity, sleep and heart rate but lacks demographic information and the number of unique users varies across tables, indicating incomplete data coverage.
- **Current:** Limited – data from 2016 may not reflect current user behaviour
- **Cited:** Yes – publicly available on Kaggle with clear licence

Limitations

The dataset has a relatively small sample size and lacks demographic data such as gender, age and location. Additionally, variation in user counts across tables suggests partial data availability. Since Bellabeat primarily targets women, these limitations mean that the findings should be treated directional rather than definitive.

Process – Data Cleaning and Preparation

Tools Used

Tool	Purpose
Microsoft Excel	Data cleaning, transformation and preparation
SQL(BigQuery and MySQL)	Data querying, filtering and aggregation; BigQuery for general analysis, MySQL Workbench for large-scale heart rate data analysis
Command-line tools	Handling and merging large CSV files (~3.6 M records)
Tableau	Data visualization and dashboard creation

Data Cleaning Steps

1. Removed duplicate records to ensure each user-day entry was unique
2. Handled missing values by removing or excluding incomplete rows
3. Standardised date and time formats for consistent analysis across datasets
4. Verified data types for all columns (numeric, date, text)
5. Created derived columns (e.g., total steps, total activity minutes, activity level category, day of week and hourly time segments) to support deeper analysis.
6. Standardized column names by converting camelCase to snake_case using Excel formulas (LOWER) and ensuring consistent formatting with underscores.
7. Merged daily activity, sleep and heart rate datasets using User ID and Date as keys

Feature Engineering (New Variables Created)

- Activity Level Classification: Users were categorised into activity levels(Sedentary, Lightly Active, Active, Moderately Active and Highly Active) based on total step count
- Night Heart Rate: Extracted average heart rate during night hours (12 AM to 6 AM) for recovery analysis
- Peak Activity Window: Identified 4 to 7 PM as the peak daily activity period based on average hourly step and heart rate patterns

Additional Data Processing – Heart Rate Analysis

1. Processed and analysed approximately **3.6 million heart rate records** to extract meaningful insights
2. Used **command-line tools** to efficiently merge and manage multiple large CSV files
3. Performed data querying and analysis using MySQL Workbench
4. Created aggregated tables to summarise heart rate trends for analysis.

These processing and transformation steps ensured the dataset was structured, enriched, and ready for meaningful analysis.

Analyze – Data Analysis and Key Insights

Objective of Analysis

The goal of this analysis is to identify patterns in smart device usage and understand user behaviour across physical activity, sleep, calories burned and heart rate metrics. These insights will help Bellabeat improve its marketing strategy and product engagement.

Key Metrics Summary

Metric	Value	Context
Average Daily Steps	7,281	Below 10,000 daily target
Average Sleep Duration	7.0 hrs	Below 7-9 hr recommendation
Average Night Heart Rate	64 bpm	Within normal resting range

Analysis and Key Findings

The analysis reveals important behavioural patterns in user activity, health, and recovery, providing valuable insights for product and business strategy.

1. Activity Patterns

User activity levels are below the recommended daily 10,000-step benchmarks, with an average of 7,281 steps. Activity levels vary across the week, peaking on Saturdays and dropping to their lowest on Sundays. Additionally, 41.3% of users fall into the 'Lightly Active" category.

→ **Insight:** The findings reveal an opportunity to enhance user engagement by promoting more consistent daily activity, particularly during low-activity days such as Sundays.

2. Steps vs Calories Burned

The moderate positive correlation ($R^2 = 0.37$) confirms that increased physical activity contributes to higher calorie expenditure, although other factors (such as metabolism or workout intensity) also play a role.

→ **Insight:** Step tracking is a meaningful but incomplete indicator of overall fitness, suggesting the need for more holistic activity tracking.

3. Heart Rate Patterns

Analysis of 3.6 million heart rate records identified a consistent peak in user heart rates between 4 PM and 7 PM, indicating this as the primary activity window.

→**Insight:** This represents an optimal engagement window where users are most physically active and more likely to respond to fitness prompts or challenges.

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4. Sleep and Recovery

The strong negative correlation ($R^2 = 0.66$) between night heart rate and sleep duration suggests that elevated physiological stress or poor recovery is associated with reduced sleep quality.

→ **Insight:** Elevated physiological stress or poor recovery may be negatively impacting sleep quality, making it a critical focus area for improving user well-being.

Summary

The analysis highlights three key behavioural trends:

- **Suboptimal activity levels**
- **Concentrated activity during evening hours**
- **Strong relationship between recovery and sleep quality**

The findings suggest that while users exhibit moderate engagement, there is significant potential to improve activity consistency, capitalize on peak engagement windows, and enhance recovery – enabling more effective and personalized health interventions.

Share – Data Visualizations and Dashboard

Key Visualizations

Three interactive Tableau dashboards were created to communicate findings to stakeholders:

Dashboard 1 – User Activity & Behaviour Overview

Provides a summary of user activity trends, highlighting step distribution, weekly engagement patterns and the relationship between physical activity, calorie burn and sleep behaviour.

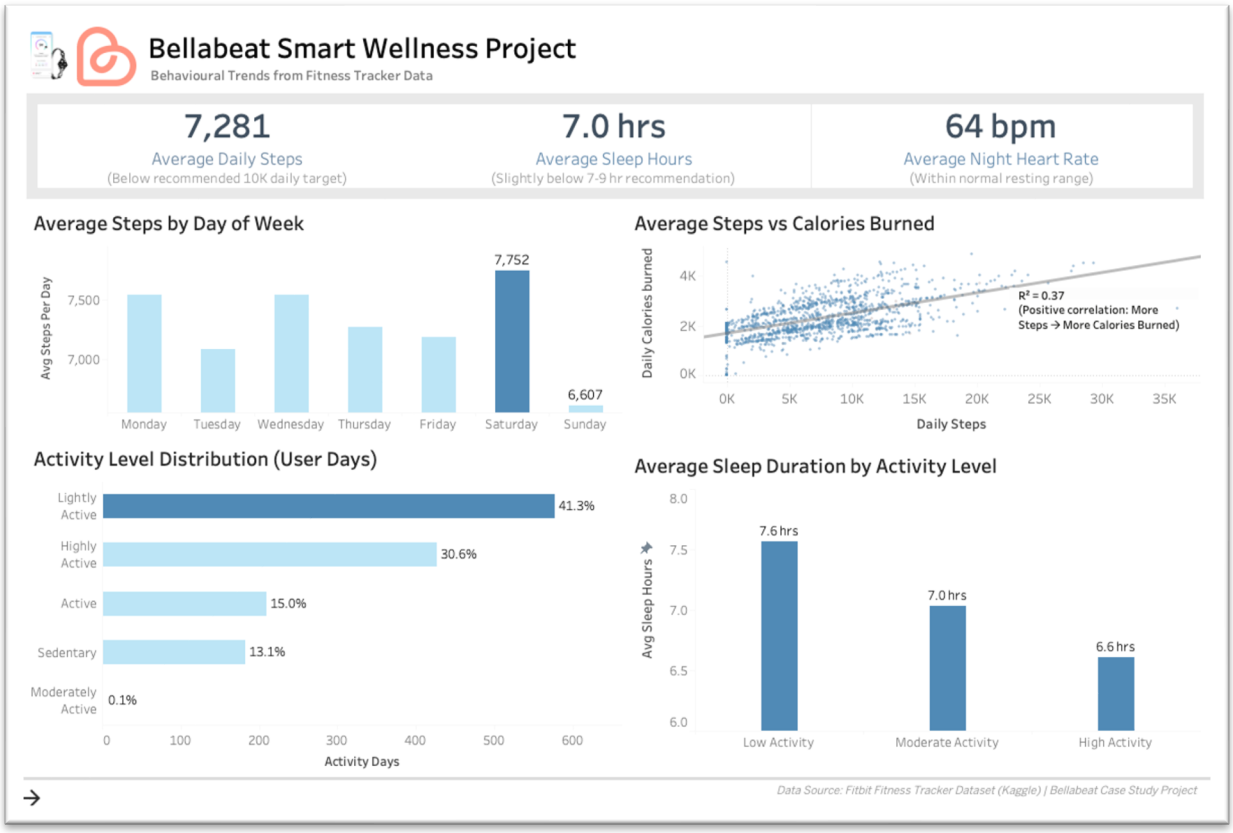

Figure 1: Bellabeat Smart Wellness Project – Overview

These visuals reveal that users consistently fall short of recommended activity targets. Engagement peaks on Saturdays and drops on Sundays – highlighting an opportunity for targeted strategies to boost weekend activity.

Dashboard 2 – Heart Rate & Recovery Insights

This dashboard analyzes key recovery patterns by examining heart rate behaviour, peak engagement periods and their impact on sleep quality.

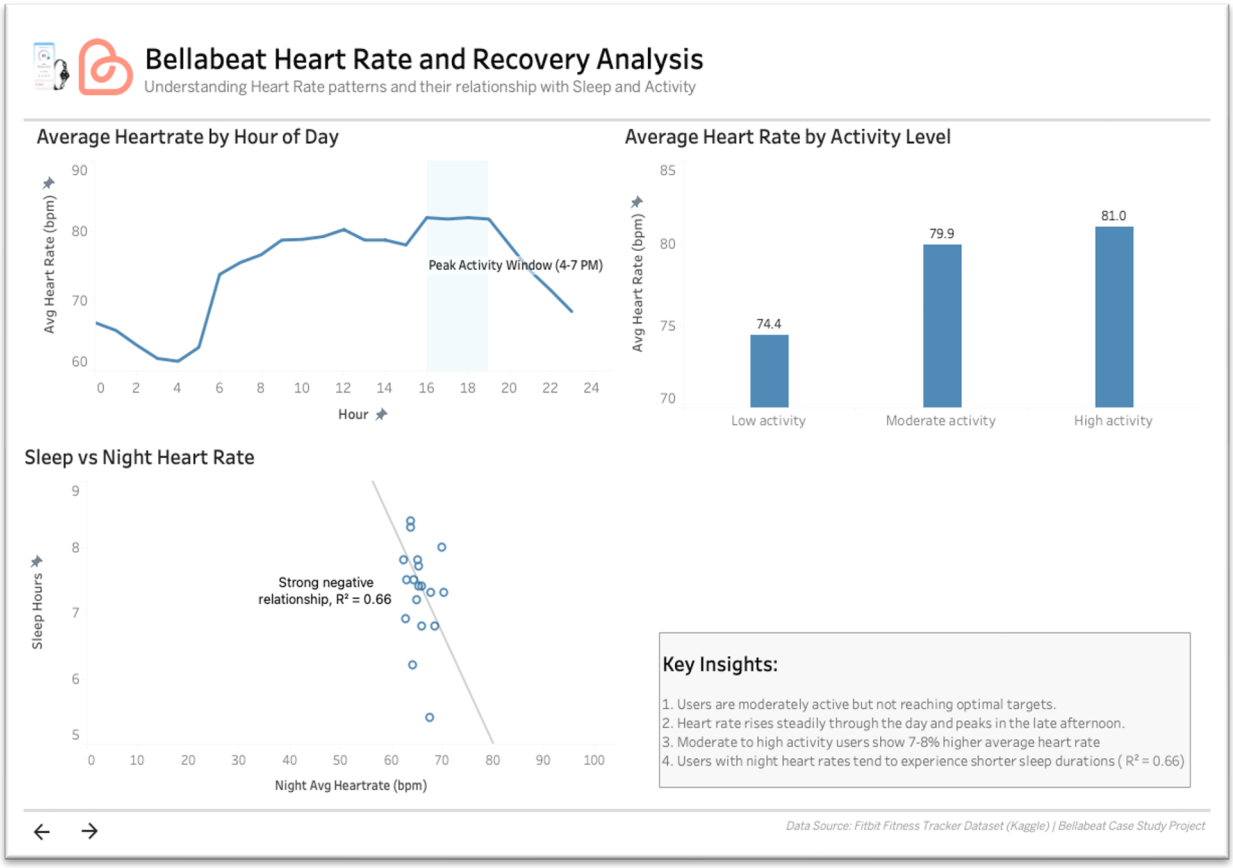

Figure 2: Heart Rate and Recovery Insights

Users are most active during the evening (4–7 PM), while elevated night heart rates are associated with reduced sleep duration, highlighting a key opportunity to improve recovery and overall well-being.

Dashboard 3 – Business Recommendations and Impact

This dashboard presents strategic recommendations based on user activity and recovery insights, along with their potential impact on engagement, retention and subscription growth.

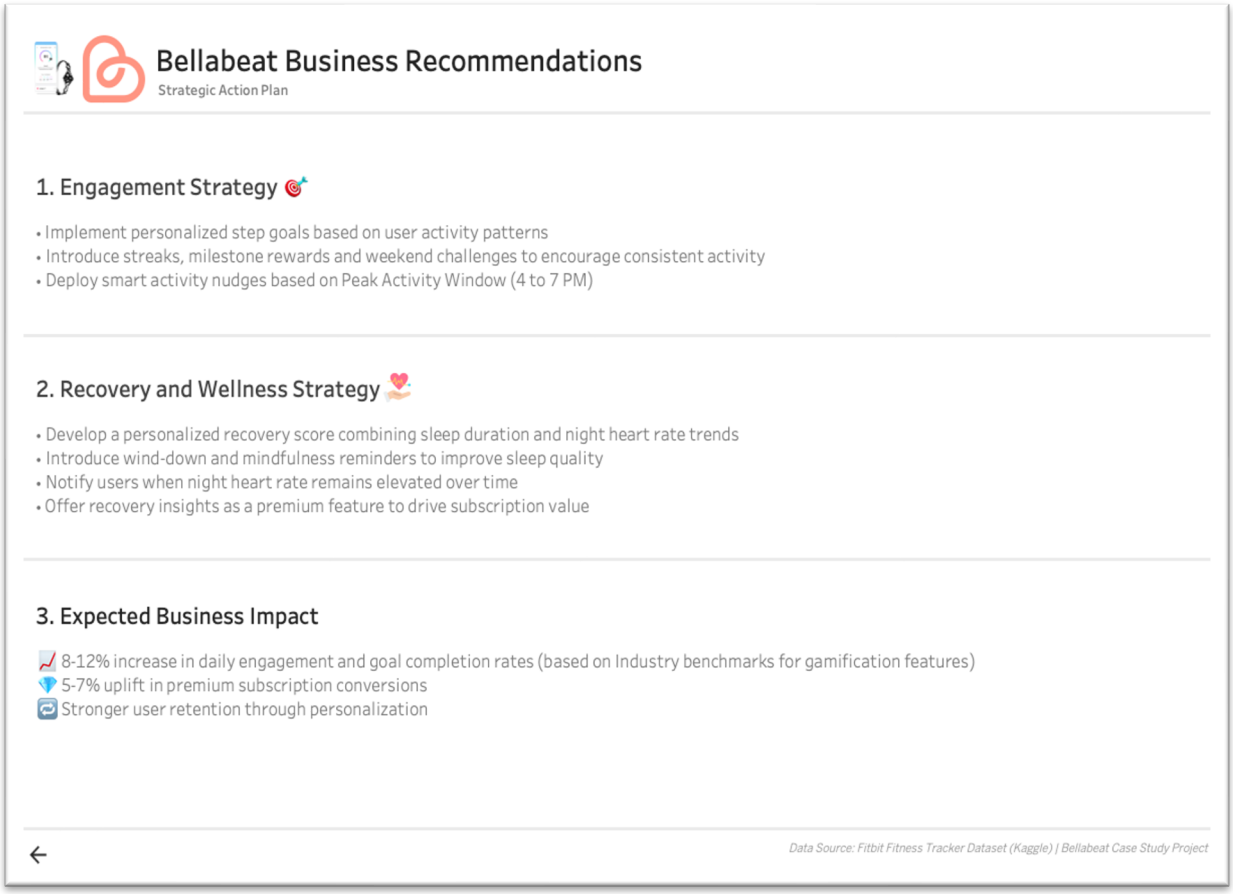

Figure 3: Business recommendations and impact

Dashboard 3 demonstrates how analytical insights can be translated into **actionable product strategies** that enhance user experience while driving measurable business growth.

Summary

These insights served as the foundation for formulating data-driven business recommendations in the final phase.

Act – Business Recommendations

Recommendation 1 – Engagement Strategy

- Implement personalised step goals based on individual user activity patterns rather than a one-size-fits-all 10K target
- Introduce streaks, milestone rewards, and weekend challenges to encourage consistent activity, particularly on Sundays which showed the lowest step counts
- Deploy smart activity nudges timed to the Peak Activity Window (4 to 7 PM) when users are most likely to be active

Recommendation 2 – Recovery and Wellness Strategy

- Develop a personalised Recovery Score combining sleep duration and night heart rate trends to give users a holistic wellness metric
- Introduce wind-down and mindfulness reminders in the evening to improve sleep quality, particularly for highly active users who showed shorter sleep durations
- Notify users when night heart rate remains elevated over consecutive days, which may signal insufficient recovery
- Offer Recovery Insights as a premium feature to drive subscription value and differentiate Bellabeat in the market

Expected Business Impact

Outcome	Projected Impact
Daily Engagement & Goal Completion	8-12% increase (based on industry benchmarks for gamification features)
Premium Subscription Conversions	5-7% uplift through recovery insights as a premium feature
User Retention	Stronger retention through personalisation and recovery-focused features

Limitations and Next Steps

Limitations

- The dataset is limited to **35 users from 2016**, which may not fully represent current user behaviour.
  A larger and more recent dataset would strengthen the reliability of the insights.
- The **number of unique users varies across different tables**, indicating incomplete data coverage and potential inconsistencies during analysis.
- **Demographic data** such as **age, gender, location** are not available, limiting the ability to create highly targeted and personalized recommendations.
- The dataset does not include **women-specific health metrics** (**menstrual cycle tracking**), which are integral to Bellabeat's product offering, and may not fully capture real-world user behaviours such as device non-usage, potentially impacting data accuracy.

Next Steps

- Conduct **A/B testing of proposed features** (e.g., gamification, personalized goals) to measure impact on user engagement and retention.
- Incorporate **Bellabeat's proprietary user data** for more accurate, real-time and business-relevant insights.
- Continuously monitor key metrics such as **daily active users, engagement rates and goal completion rates** to refine and optimize product features.
- Implement a feedback loop to continuously collect user input and improve feature effectiveness.